CHAPTER 14

PROBABILITY

CHAPTER ESSENCE AT A GLANCE:
Probability is the mathematics of chance! Total probability of all possible outcomes in sample space S always equals 1. If two events can happen together, subtract their overlap P(A B) so you don't double count!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Sample Space & Events:** Sample space S is the set of all possible outcomes. Event E is a subset of S. Classical probability: P(E) = n(E) over n(S). Always 0 ≤ P(E) ≤ 1.
- **Addition Theorem of Probability:** P(A B) = P(A) + P(B) - P(A B). For mutually exclusive events (A B =): P(A B) = P(A) + P(B).
- **Complementary Events:** P(A') = P(not A) = 1 - P(A). Probability that NEITHER occurs: P(A' B') = P((A B)') = 1 - P(A B).
- **At Least One Probability Shortcut:** P(at least one) = 1 - P(none). Solves multi-card, dice, and exam pass problems in 5 seconds!

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$P(E) = \frac{n(E)}{n(S)} \quad (0 \leq P(E) \leq 1)$$

Classical Axiomatic Probability Definition

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

General Addition Theorem of Probability for Two Events

$$A \cap B = \emptyset \rightarrow P(A \cup B) = P(A) + P(B) \quad \text{(Mutually Exclusive Events)}$$

Special Addition Rule for Mutually Disjoint Events

$$P(A') = 1 - P(A), \quad P(A \cap B') = P(A) - P(A \cap B)$$

Complementary Event and Difference Probability Axioms

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Probability of any event MUST lie between 0 and 1 inclusive (0 ≤ P(E) ≤ 1). If your answer is negative or greater than 1, check your math!
- **Alert:** Whenever a question asks for 'Probability of AT LEAST ONE', calculate 1 - P(None) for a fast 10-second solution!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. A card is drawn from a well-shuffled deck of 52 cards. What is the probability that the card is: (a) A diamond? (b) An ace? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: (a) Diamond Probability]** There are 13 diamond cards: P(Diamond) = 13 over 52 = 1 over 4. [1 Mark]

• **[Step 2: (b) Ace Probability]** There are 4 aces in the deck: P(Ace) = 4 over 52 = 1 over 13. [1 Mark]

• **[Boxed Result: (a) 1 over 4, (b) 1 over 13]**

2. Given P(A) = 0.54, P(B) = 0.69, and P(A intersection B) = 0.35. Find: (a) P(A union B) (b) P(A'). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: (a) P(A union B)]** P(A union B) = P(A) + P(B) - P(A intersection B) = 0.54 + 0.69 - 0.35 = 1.23 - 0.35 = 0.88. [1 Mark]

• **[Step 2: (b) P(A')]** P(A') = 1 - P(A) = 1 - 0.54 = 0.46. [1 Mark]

• **[Boxed Result: (a) P(A U B) = 0.88, (b) P(A') = 0.46]**

3. If E and F are events such that P(E) = 1 over 4, P(F) = 1 over 2, and P(E and F) = 1 over 8, find: (a) P(E or F) (b) P(not E and not F). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: P(E or F) = P(E union F)]** P(E union F) = P(E) + P(F) - P(E intersection F) = 1 over 4 + 1 over 2 - 1 over 8 = 2 over 8 + 4 over 8 - 1 over 8 = 5 over 8. [1.5 Marks]

• **[Step 2: P(not E and not F)]** By De Morgan's Law: E' intersection F' = (E union F)'. P(not E and not F) = 1 - P(E union F) = 1 - 5 over 8 = 3 over 8. [1.5 Marks]

• **[Boxed Result: (a) P(E or F) = 5 over 8, (b) P(neither) = 3 over 8]**

4. A box contains 10 red marbles, 20 blue marbles and 30 green marbles. 5 marbles are drawn from the box, what is the probability that: (a) All will be blue? (b) At least one will be green? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Total Combinations]** Total marbles = 10 + 20 + 30 = 60. Ways of choosing 5 marbles = 60C5. [0.5 Mark]

• **[Step 2: (a) All 5 Blue]** Blue marbles = 20. Favorable ways = 20C5. P(All Blue) = 20C5 over 60C5. [1 Mark]

• **[Step 3: (b) At Least One Green]** Non-green marbles = 10 + 20 = 30. Ways of choosing 5 with no green = 30C5. P(At least one green) = 1 - P(No green) = 1 - (30C5 over 60C5). [1.5 Marks]

• **[Boxed Result: (a) 20C5 over 60C5, (b) 1 - (30C5 over 60C5)]**

5. In a lottery, a person chooses six different natural numbers at random from 1 to 20, and if these six numbers match with the six numbers already fixed by the lottery committee, he wins the prize. What is the probability of winning the prize in the game? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Total Outcomes]** Total ways to choose 6 numbers from 20 = 20C6. 20C6 = (20 × 19 × 18 × 17 × 16 × 15) over (6 × 5 × 4 × 3 × 2 × 1) = 38,760. [1.5 Marks]

• **[Step 2: Winning Probability]** There is only 1 winning ticket matching all 6 numbers. P(Winning) = 1 over 20C6 = 1 over 38,760. [1.5 Marks]

• **[Boxed Result: Probability = 1 over 38,760]**

6. (a) State and prove the Addition Theorem of Probability for any two events A and B. (2.5 Marks)n(b) If P(A) = 0.5, P(B) = 0.3, and P(A intersection B) = 0.15, find P(A' intersection B'). (1.5 Marks) [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Addition Theorem Proof]** By set theory: n(A U B) = n(A) + n(B) - n(A n B). Dividing both sides by total outcomes n(S): n(A U B) over n(S) = n(A) over n(S) + n(B) over n(S) - n(A n B) over n(S) ⇒ P(A U B) = P(A) + P(B) - P(A n B). If A and B are mutually exclusive, P(A n B) = 0 ⇒ P(A U B) = P(A) + P(B). [2.5 Marks]

• **[Step 2: Numerical Calculation]** P(A U B) = 0.5 + 0.3 - 0.15 = 0.65. By De Morgan's Law: P(A' n B') = P((A U B)') = 1 - P(A U B) = 1 - 0.65 = 0.35. [1.5 Marks]

• **[Boxed Result: P(A U B) = P(A) + P(B) - P(A n B) proved; P(A' n B') = 0.35]**

7. In a class of 60 students, 30 opted for NCC, 32 opted for NSS and 24 opted for both NCC and NSS. If one of these students is selected at random, find the probability that: (a) The student opted for NCC or NSS (b) The student has opted for neither NCC nor NSS (c) The student has opted for NSS but not NCC. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: (a) NCC or NSS]** P(A) = 30 over 60, P(B) = 32 over 60, P(A n B) = 24 over 60. P(A U B) = P(A) + P(B) - P(A n B) = 30 over 60 + 32 over 60 - 24 over 60 = 38 over 60 = 19 over 30. [1.5 Marks]

• **[Step 2: (b) Neither NCC nor NSS]** P(Neither) = 1 - P(A U B) = 1 - 19 over 30 = 11 over 30. [1 Mark]

• **[Step 3: (c) NSS but not NCC]** P(B - A) = P(B) - P(A n B) = 32 over 60 - 24 over 60 = 8 over 60 = 2 over 15. [1.5 Marks]

• **[Boxed Result: (a) 19 over 30, (b) 11 over 30, (c) 2 over 15]**